

Topic A Part 2 Review [108 marks]

1. [Maximum mark: 20]

24N.2.HL.TZ0.4

The police are considering changing from a relational database to either a network database or an object-oriented database.

(a) Identify **two** characteristics of a network database.

[2]

Markscheme

Award [2 max]

Hierarchical Structure/Tree-like structure;
Multiple parent records/Multiple children records;
Data linked to other data points/web-like structure of nodes;
Natural relationships/relationship navigation;
Many to many relationships between entities;
Flexible data access due to the network of relationships;

Note: Accept an example, criminal linked to crime records.

Examiners report

Many students were unable to identify valid characteristics of a network database. A good number of them provided responses that relate to a computer network, which is off course.

(b) Discuss the advantages **and** disadvantages of an object-oriented database compared to a relational database.

[5]

Award [5 max]

Advantages

OODBS allow efficient representation and management of complex data structures and relationships e.g. they allow the storage of objects multimedia data/polygons/points as opposed to storing integers, strings and real numbers in RDBMS;

OODBS uses similar features to OOP (inheritance, encapsulation, polymorphism) which are unavailable to RDMS. These features support code reusability and simplify application development;

OODBS offer improved performance as they can directly store program objects. RDBMS must first flatten program objects before storing and recreate objects when loading them. This creates performance impedance; OODBS are better at modelling real-world objects and relationships than RDBMS. It is easy to extend functionality of objects by modifying the underlying classes through inheritance and polymorphism;

Disadvantages

RDBDS is more widely used/more accessible so there is a lot for support for the database model;

The availability of numerous vendors of RDBMS means there cannot be vendor-lock in as in OODBS;

RDBMS has greater compatibility (interoperability) with other systems;

RDMS are easy to design and modify and extend/OODBS are more complex to modify and extend;

RDMS have standard query tools/report generators;

RDMS has higher integrity and lower redundancy due to availability of various constraints like referential integrity constraints, domain integrity constraints;

RDMS offer better performance in heavy multiuser systems because the data model handles concurrency better and more effectively than OODMS;

RDMS are better at dealing with large datasets/complex queries;

Conclusion/Opinion

The conclusion should do more than repeat the stated information.

Marks [2] + [2] + [1]

Examiners report

This was generally well-answered by the average student. Most of them, however, failed to include a conclusion as required by the command term.

The European Union Agency for Law Enforcement Cooperation (Europol) may develop a data warehouse to hold data related to crimes.

- (c) Outline **two** advantages of creating a data warehouse to store crime data.

[4]

Markscheme

Award [4 max]

Centralises data from multiple sources;
Cleans the data through the ETL processes;
So that standardised searches can be used/to avoid separate searches on different sources;

Data warehouses have more data than a database/petabytes or terabytes compared to gigabytes;
Which increases the chances of finding patterns/noticing anomalies;

Databases only store transaction data/day to day operations;
Data warehouse time stamps data so that when the data was collected can

be factored into the analysis;

Providing historical intelligence (and making trend analytics possible);

Data warehouses have advance OLAP tools;

Which allows data mining/data matching/link analysis techniques/allows for classification, cluster analysis, and associations/allows for predictions;

Marks [2] + [2]

Examiners report

Majority of students provided good reasons for using a data warehouse to store crime data. A good number of students lost marks because their responses did not meet the demands of the command term.

Predictive modelling is a technique that could be applied to the data warehouse.

- (d) Outline **two** benefits of applying predictive modelling to crime data.

[4]

Markscheme

Award [4 max]

PM is an accurate way to predict and forecast likely future crimes;

By analysing demographics/socio-economic indicators/environmental factors;

PM outcomes enable Europol to allocate resources/resource optimization;

Europol has limited resources, so PM makes the best use of them;

PM will automatically update to factor in changes in the underlying data;
With an emphasis on information-bearing patterns;

PM will ignore random patterns/outliers/errors/variation (noise) so they
don't affect outcomes;

And can be applied to structured and unstructured data;

PM often leverages domain-specific knowledge or rules;

Making it useful when patterns are well-defined like cases on homicide;

PM can unearth crime patterns linked to specific times and events;

Enabling law enforcement agencies to develop strategies and be ready in
those periods.

Marks [2] + [2]

Examiners report

Many students were able to outline at least one reason for using predictive modelling but struggled to obtain all the marks. Again, many students lost marks because their responses did not meet the demands of the command term.

Europol would like to apply link analysis to investigate fraud, uncover terrorist networks and investigate criminal activities.

(e) Explain how link analysis can be applied to crime datasets.

[5]

Markscheme

Award [5 max]

Link analysis identifies relationships between large quantities of crime data;
Link analysis examines relationships among individuals, locations, organizations, objects and events;
Link analysis can be applied to many databases (associating data between them);
Link analysis may discover new patterns of interest (including individuals, locations, organizations, objects and events);
Link analysis can check the similarity between the datasets using data matching techniques;
Link analysis can find anomalies where known patterns are violated;
Identified anomalies can be linked to fraudulent activities/behaviours requiring further investigation;
Accept example, such as criminal network using phone records/credit card theft identification using patterns of transactions;

Note: Award 4 marks max if there is no reference to the scenario

Examiners report

This was not particularly well-answered. Most students demonstrated understanding of link analysis on its own but they were largely unable to coherently explain how it can be applied to crime datasets.

2. [Maximum mark: 18] 24M.2.SL.TZ0.2

The Bucharesti School website allows parents to login and select school transportation for their children. If they select the school bus, they will have to pay for this service at the end of the month.

- (a) Identify the steps that take place in a transaction when a parent attempts to pay for the school bus at the end of a month.

[3]

Markscheme

Award [3 max].

Parent authenticated by the DBMS;
Outstanding amount/bill is calculated/displayed;
Transaction is initiated for the transport payment;
Payment details added / entered for transport;
If payment details and payment can be processed by DBMS, then School transport Account is credited;
... parent's account is debited;
Transaction is completed by DBMS successfully//The transaction is committed;
Automatic e-mail notification of payment sent to parent;
Else, Transaction is rolled back;
Notification of the failed/successful transaction sent to the parent;

Examiners report

Most students were able to outline the required steps in a logical manner. Others wrote a lot about the properties of transactions without relating the properties to the scenario with specifics.

- (b) Explain how the database management system (DBMS) prevents a record being updated by two parents simultaneously.

[3]

Markscheme

Award [3 max].

Record locking /Isolation/Data Locking/Row locking;

Ensures exclusive editing/it is done in isolation;

... to prevent the same data item from being changed by two different transactions / to prevent data values from being accessed by a transaction while another transaction involving the same data is in progress / the process helps keep the update mutually exclusive;

Optimistic Concurrency Control (OCC) // Multi-version concurrency control (MVCC);

Allows multiple users to access the unmodified version of data at the same time // Uses read isolation;

....To prevent dirty reads / reading the uncommitted changes of another transaction;

On update request, a check is done to see if the existing data has been modified by another user since it was initially read // Uses write validation;

....To prevent lost updates / overwriting changes from other transactions;

Note: The question asks for an explanation, so marks can be awarded from both blocks.

Examiners report

A large majority of students explained row-locking adequately, but they could not proffer any other method to enable them to get all the marks. Only a handful of students referred to Optimistic Concurrency Control or Multi-version concurrency control.

The students' personal details must be securely stored in the database.

- (c) Identify **two** roles of the database administrator at Bucharesti School.

[2]

Markscheme

Award [2 max].

Approving Data Access // Managing user accounts;
Monitoring Performance // Performance tuning;
Backup and Recovery;
Implementing security;
Upgrading/updating the database // Maintenance;

Examiners report

This was generally well-answered. Though the question only required students to identify the roles of a DBA, some produced lengthy responses, wasting time.

- (d) Outline **two** ways that a database management system (DBMS) can be used to ensure the students' personal data remains secure.

[4]

Markscheme

Award [4 max].

Access controls;
Effective and efficient access controls/restrictions so that the end users can access only the data or programs for which they have legitimate privileges;

User accounts;
Different users have usernames and passwords/biometrics to enable unique login experiences;

Data Encryption;
Nullifies the potential value of data interception;

Ensures the confidentiality of data;

*Do **not** credit answers that refer to data integrity rather than security.*

Marks as [2] and [2]

Examiners report

The majority of the students answered the question well. A few of them lost marks because they mixed up authentication and authorisation: they would indicate the use of logins and then give the outline of authorization rather than authentication.

- (e) Explain how the developers of the Bucharesti School database can ensure that it has been designed ethically.

[6]

Markscheme

Award [6 max].

Privacy considerations must be ensured
Prevention of unauthorized access to private data//providing security measures e.g. logins;
Encryption of data to ensure it is unusable to unauthorized users;
Especially as many of the data subjects will be under 18;
Ensuring that the inappropriate use of data cannot take place, for example sharing the data with third parties without the consent of data subjects;
Measures to ensure accuracy and completeness when collecting data about a person or persons (corporations included) by technology;
Availability of data content, and the data subject's legal right to access;
Ownership rights to inspect, update or correct these data;
The data is not available to the vast majority of its users //Views are used to

access specific data instead of giving access to the whole table //Redaction;
The data base is designed to conform with data protection legislation;
Data must not be kept for longer than it is required;
Only relevant data must be collected and stored;
Keeping the data secure from loss or damage e.g. backup;

Examiners report

The majority of students found marks hard to come by in this question because they concentrated on the ethical use of the database instead of how to ensure that the database has been designed ethically.

3. [Maximum mark: 20]

24M.2.HL.TZ0.4

Books are sold in physical bookshops and by online retailers. Each retailer maintains a unique database of books sold.

Online retailers can have a much greater range of books available. Books that are very rarely sold are not stocked and can be printed on demand.

Information from each retailer's database is loaded into a data warehouse where data analytics take place.

(a) Outline **one** advantage of using a data warehouse.

[2]

Markscheme

Award [2 max].

They enhance business intelligence;
Since they integrate huge amounts of data from different sources;
Managers and executives can make more informed decisions based on **diverse** data;

They reduce decision-making time //Timely Access to data;
Users can quickly access critical data all in one place // they won't waste time to access data from multiple sources;
Executives can query data/generate reports themselves without waiting for technical support;

Enhances Data Quality and Consistency;
Data transformation / Standardisation/ cleansing is enforced;
Ensuring data accuracy and consistency across different sources // eliminating inconsistencies in data from different sources;

Provides Historical Intelligence;
They store large amounts of historical data that allow analysis of different time periods and trends;
In order to make future predictions;

Provides more efficient access to data // Improves query processing time;
As data is optimised for reporting (rather than transaction processing);
Through de-normalization / removal of constraints;

Note: Award any **two** from **one** group only.

Examiners report

This was the best-answered question, with most students providing excellent reasons for using a data warehouse. A few students, however, demonstrated a complete lack of understanding of the concept by referring to data warehouses as extensions of transactional databases that allowed larger amounts of data to be stored.

- (b) Explain why the data from the individual retailers is transformed before it is loaded into a data warehouse.

[3]

Markscheme

Award [3 max].

When the data is collected from different book retailers, each will have their own formats or standards;

Data must be standardized to conform with the structure of the data warehouse;

The data is transformed to select the data that is useful for analysis;

The data is optimised for retrieval;

Aggregation of data // summarizing and consolidating large sets of detailed data into a more concise and manageable data set;

...Improves query performances and reduces storage requirements;

//Allow an example related to the scenario (book sales) such as splitting dates into day, month, year for book sales records;

Examiners report

A good number of students showed an understanding of the transformation process and identified some activities carried out, but the reason for the transformation was missing in most cases. A number of students also failed to relate their responses to the scenario.

(c) Explain why data warehouses use timestamping.

[3]

Markscheme

Award [3 max].

Data warehouse update happens in real time // Data in a data warehouse is time dependent // Data is only valid for a period;

The user's needs (data required) change with time;
A record update on a transaction is unique with the timestamp;
Since a data warehouse contains both current and historical data;
... timestamps are required to compare data from different times // ... helps to know the last update of a record;
The Retention Limitation Obligation prohibits organisations from holding personal data indefinitely if there are no legal or business justifications for doing so;
... therefore, timestamping can be used as a tool to help ensure data is current;

Examiners report

This question was not particularly well-answered. The more able students gave very good reasons for the use of timestamping. A large majority of the students could only identify knowledge of the time of data loading into the data warehouse as a use for timestamping.

Predictive modelling uses decision trees and neural networks.

- (d) Distinguish between decision trees and neural network learning algorithms.

[2]

Markscheme

Award [2 max].

construction of **decision tree** classifiers does not require any domain knowledge or parameter setting, and therefore appropriate for exploratory knowledge discovery;

May struggle to capture complex, non-linear relationships between features. Generally easy to assimilate/understand by humans // Easy to understand the logic behind the decision-making process;

The learning and classification steps of decision tree induction are simple and fast;

Decision tree classifiers have good accuracy with smaller sets of structured/tabular data;

Decision tree induction algorithms have been used for classification;

Uses estimates and probabilities to calculate likely outcomes;

Neural Networks learning algorithms:

Needs domain knowledge and parameter setting;

More complex to assimilate/understand by humans;

Superior for tasks involving complex patterns and high-dimensional data (e.g., image and speech recognition);

High tolerance of noisy data as well as their ability to classify patterns on which they have not been trained;

Requires significant computational resources and time;

Potential for high accuracy with large datasets;

Uses activation functions and link weights to calculate the next output value;

Note: Award **one** mark for decision tree **and one** mark for neural network learning algorithms. Do not give both marks for giving the opposite answer to one already given.

Examiners report

This was not well-answered, as most of the students concentrated on the structural differences between decision trees and neural networks instead of the differences between their algorithms.

Different data mining techniques are used to discover patterns.

(e) Outline why deviation detection is used in data analytics.

[4]

Markscheme

Award [4 max].

To provide a statistical technique/method;

... to direct appropriate marketing to certain types of books;

... which helps to detect outlying data that does not fit the assumed model;

For example, to detect books with abnormally high or low sales;

It can be used to predict the trends and patterns of demand for certain books/genres in the future;

It can help discover new opportunities for product development and marketing strategies;

As anomalies can sometimes reveal hidden patterns or trends that weren't previously known;

For example, very high book sales in a particular month can reveal new market niche for that book, informing increasing stock to meet demand for the period;

Detecting anomalies can point to the presence of errors or fraud in transactions;

Further investigation can be done to fix the error or detect/prevent the fraud;

Note: Max 3 if **no** link to the scenario is seen.

Examiners report

Most students demonstrated a good understanding of data deviation but struggled to apply it appropriately to the scenario. Successful students supported their explanation with examples based on the scenario.

- (f) Compare cluster analysis and forecasting as techniques to understand and predict data in data mining.

[6]

Markscheme

Award [6 max].

Cluster analysis employs unsupervised learning strategy // there are no pre-existing data labels;

To discover groups/clusters of data points based on some similarities/differences within the data;

The main focus is to find hidden patterns in the data points and segment them for further analysis;

It is applied in anomaly detection, target marketing, fraud detection analysis, market segmentation etc. //accept **one** example relating to the scenario;

Forecasting is a supervised learning strategy // usually uses historical data with labelled target variables;

To predict (future) values based on historical data //To predict the value of a dependent variable based on the value(s) of independent variable(s);

The main focus is predicting (future) trends or events;

It is applied in call volume prediction, weather forecasting, demand forecasting, stock market prediction etc. //accept **one** example relating to the scenario;

Marks [3] and [3]

Examiners report

Many students demonstrated an understanding of cluster analysis and forecasting individually as data mining techniques, but most struggled to

adequately compare them based on their uses.

4. [Maximum mark: 15]

23N.2.SL.TZ0.2

The restaurant owner has hired a database administrator to extend the database.

The database administrator has decided to delete the LinePrice field from the ORDERLINE table.

ORDERLINE

<u>OrderID</u>	<u>ItemID</u>	Quantity	UnitPrice	LinePrice
DM1007	F1234	1	16.99	16.99
DM1007	W2134	2	6.99	13.98

- (a) Explain why the decision to delete the LinePrice field has been made.

[3]

Markscheme

Award [3 max]

The user may type in the wrong amount;

LinePrice should be a derived field/calculated field $= [\text{Quantity}] * [\text{UnitPrice}]$;

It is not necessary to store this data/wastes storage space;

The relation/table does not conform to 3NF;

Examiners report

Most students observed that **LinePrice** depends on the **Quantity** and the **UnitPrice as a calculated/derived field**; some of them identified the waste of storage space as a reason for deleting the field. Only the more able students referred to normalization to 3NF as a reason.

The database administrator has created new tables to store employee data and create a work schedule.

When an employee works a shift in the restaurant, their hours and pay rate are recorded by the shift manager as a database transaction.

(b) Define the term *database transaction*.

[1]

Markscheme

Award [1 max]

A logical unit of work executed in full or not at all;
A change in the database state;

Examiners report

A good number of students were able to define a database transaction but too many of them showed vague to trivial understanding of the concept.

(c.i) State **one** way to validate the hours_worked field.

[1]

Markscheme

Award [1 max]

Range check/accept example, >0 and <=12;
Presence check/a blank field may cause a calculation error;
Type check/must enter a number/ float/integer;

Examiners report

A good number of students responded appropriately but too many of them showed a lack of understanding of validation and the different types of data validation check.

(c.ii) Outline **one** way to verify the payrate field.

[2]

Markscheme

Award [2 max]

Proofreading/Comparing the payrate to hardcopy version;
Lookup payrate in another table;
Based on the selected job/jobID;
double data entry/Two-pass verification;

Examiners report

The majority of students showed a complete lack of understanding of the concept of data verification. Some described validation checks instead.

The shift manager sometimes changes the employees' work hours in the database during their shift. Employees can access the work hours and confirm their availability.

(d) Explain how data integrity can be maintained if an employee tries to access their work hours while the shift manager is editing them.

[4]

Markscheme

Award [4 max]

Only allow waiters to view work hours not approve them during a shift;

Writing to the table could only be open at the end of the shift/set times;
The database could be set to lock the record/table while it is being edited;
So the waiter could not approve the hours while they are edited;
The database could be set to block viewing while the table is being edited;
A message could be conveyed to the waiter that the data is being updated;
Two-phase locking can be applied to guarantee full isolation of the update;
Multi-version concurrency control (MVCC) could be used to provide the waiter the prior, unmodified version of data that is being modified;

Examiners report

A good number of students identified row/table locking as a method the concurrency control system could use to ensure the employees do not access the data while the shift manager is making changes. A considerable number of students were vague in the method used; they insufficiently indicated that the concurrency control system would not allow it.

The database administrator has set up a procedure for database recovery.

One method of recovering a database is to use a transaction log.

- (e) Describe **one** alternative method that could be used to recover a database after a crash has occurred.

[4]

Markscheme

Award [4 max]

Instant recovery/Recovery in place;

A snapshot is created so the backup remains in a correct state;

All user write operations are redirected to that snapshot;

Users then work off the backup virtual machine (VM);

The recovery process begins in the background/Users are unaware of the recovery;

When the recovery is complete the workload is directed back to the original VM;

Crash recovery;

If a crash occurs mid-transaction;

The system follows a protocol/algorithm to recover lost data;

A check is made of the states of the transactions that were being executed;

If the transaction can be completed it should do so;

If not the transactions that were interrupted during the crash are rolled back;

No transaction can be allowed to leave in an inconsistent state;

Examiners report

The majority of students failed to consider the fact that the question specifically asked for a method to recover a database after a crash has occurred. Some of them ignored the fact that transaction log recovery is already given and could therefore not be used. Most students only gained a point for mentioning the use of a backup.

5. [Maximum mark: 15]

23N.2.SL.TZ0.3

A European pet passport allows pets to travel between countries in the European Union. All pets have a microchip inserted that gives them a unique alphanumeric transponder code.

The data contained in a pet passport is shown in **Figure 1**. All of the data is stored in a database.

Figure 1: Two pages of a pet passport

DESCRIPTION OF ANIMAL	ISSUING OF PASSPORT
PICTURE OF THE ANIMAL (optional)	Name of the authorized veterinarian: Helen Bright
Name: Benji	Address: Lowden Bridge House, Lowther Street, Eskdale
Species: Dog	Postcode: CA37 8DX
Breed: Labrador	Country: United Kingdom
Sex: Male	Telephone number: 01632 960739
Date of birth: 01/01/2017	Email address: jkr@gov.example.com
Alphanumeric transponder code: 985 AAA 111 AAA 111	Date of issue: 26/01/2019 

[Source: Department of Agriculture, Environment and Rural Affairs. EU Pet Travel Scheme - How to Complete Pet Passports. <https://www.daera-ni.gov.uk/publications/eu-pet-travel-scheme-how-complete-pet-passports>. Contains public sector information licensed under the Open Government Licence v3.0. Source adapted <https://www.daerani.gov.uk/crown-copyright>.]

A pet owner may have more than one pet, and each pet has their own passport. Pet passports are issued by registered veterinary surgeons (vets).

- (a) Construct the entity-relationship diagram (ERD) that shows the relationship between pet owners, their pets, and the pet passport.

[2]

Markscheme

Award [2 max]



Accept variations of this diagram. Owner, Pet, and Passport correctly positioned.

Award [1] for 1 to m;

Award [1] for 1 to 1;

Examiners report

The majority of students could correctly represent the entities with one-to-many and one-to-one relationships. Future students should be taught how to diagrammatically show the different relationships.

The data stored in the VET table can be represented in the following notation:

VET(VetID, FirstName, Surname, Address, Postcode, City, Country, Phone, Email)

The Phone field will be used as a secondary key.

(b) Define the term *secondary key*.

[1]

Markscheme

Award [1 max]

A secondary key is a non-identifying field used to find row in a table;

Examiners report

Most students could describe a *secondary key*.

Each year, pets require vaccinations to prevent them from contracting rabies. Information about each vaccination can be seen in the pet passport (see **Figure 2**).

Figure 2: Vaccination information page in a pet passport

VACCINATION AGAINST RABIES			
MANUFACTURER AND NAME OF VACCINE	Place vaccine sticker or handwrite: Merck Animal Health, Nobivac, 2021		
BATCH NUMBER			
1. VACCINATION DATE	1. 05/01/2019	2. 26/01/2019	3. 05/01/2021
2. VALID FROM			
3. VALID UNTIL			
AUTHORIZED VETERINARIAN	Name, address, telephone number, signature and SP number		

[Source: Department of Agriculture, Environment and Rural Affairs. EU Pet Travel Scheme - How to Complete Pet Passports. <https://www.daera-ni.gov.uk/publications/eu-pet-travel-scheme-how-complete-pet-passports>. Contains public sector information licensed under the Open Government Licence v3.0. Source adapted <https://www.daerani.gov.uk/crown-copyright>.]

The information in **Figure 1** and **Figure 2** needs to be normalized before being entered into the database.

The VET table is given and will need to be incorporated into the final structure.

- (c) Construct the database in 3rd Normal Form (3NF) for all of the entities shown in **Figure 1** and **Figure 2**.

You should use the same notation as that used in the representation of the VET table.

[8]

Markscheme

Award [8 max]

Award [3] for all 4 entities/[2] if there are only 3 entities/[1] if there are 2 correct entities.

Award [1] for correct primary keys

Award [1] for correct foreign keys

Award [1] for all correct fields in VACCINE

Award [1] for all correct fields in PETVACCINATION

Award [1] for all correct fields in PET

Award [1] for all correct fields in PASSPORT

PASSPORT(TransponderCode, VetID^(fk), DateOfIssue)

PET (TransponderCode, PetName, Species, Breed, Sex, DOB)

PETVACCINATION (VaccinationID, vaccineID, VetID^(fk), TransponderCode^(fk),

VaccineDate,

ValidFrom, ValidTo)

VACCINE (vaccineID, VaccineName, Manufacturer, BatchNumber)

Accept PassportID instead of TransponderCode in the Passport table.

Accept OwnerID in PASSPORT.

Examiners report

This was not well-answered. The majority of students identified the PET table with the correct fields and Primary Key identified. In general terms, entities were not correctly identified. Students could not identify the VACCINATION and VACCINE tables separately; they combined the fields into one table. They could not correctly identify the primary key and foreign keys for the tables. There were answers that showed that students have not understood the concept of normalization.

Access to this database must comply with data protection legislation.

- (d) Describe the principles of data protection legislation that would apply to a pet owner's data.

[4]

Markscheme

Award [4 max]

Data should only be used for the reason that it was collected;
Data should not be transferred to another country unless it has equal or better data protection law;
Data should not be shared without permission;
Data should be registered with the data protection governing body;
Data should be used in accordance with the law;
Data should be adequate and relevant;
Data should not be kept longer than necessary;
Data should be accurate and up-to-date;
The pet owner has the right to see the data stored about them;

Examiners report

This was generally not well-answered. The majority of students could not identify more than two principles of data protection legislation.

6. [Maximum mark: 20]

23N.2.HL.TZ0.4

An emergency management information system (EMIS) uses a spatial database to map all emergency calls via their coordinates. Every emergency call generates data that includes spatial data.

- (a.i) Outline what is meant by the term *spatial data*.

[2]

Markscheme

Award [2 max]

Data on the locations/shapes of geographic features;

Stored as coordinates/latitude and longitude;

Topology/properties of geometric objects/points/lines/polygons;

Examiners report

Many students could not accurately describe spatial data but they were able to say how spatial data is stored.

- (a.ii) State **two** types of geometric object that can be represented by a spatial database.

[2]

Markscheme

Award [2 max]

Lines;

Points;

Polygons;

Examiners report

Most students were able to identify at least one type of geometric object represented in a spatial database.

Spatial data on a map are always linked to a row in a table.

- (b) State **one** type of relationship that can be used to represent these links.

[1]

Markscheme

Award [1 max]

One-to-one relationships/one point is linked to one row only;

One-to-many relationships/one point is linked to many rows;

Many-to-many relationships/many points are linked to many rows;

Examiners report

About half of the students were able to identify a valid relationship. The large number of off course responses means a good number of students do not understand the concept of relationships in spatial databases.

The map in **Figure 3** shows the locations of crimes reported in Los Angeles, USA, in 2019.

The larger the dot, the more crimes that have been reported in that area.

Figure 3: Map showing the locations of crimes reported in Los Angeles, USA, in 2019



[Source: Wikipedia, n.d. *California locations by crime rate*. [online] Available at: https://en.wikipedia.org/wiki/California_locations_by_crime_rate [Accessed 11 January 2021] and Wikimedia Commons, 2008. *Locator map of the Harbor Freeway in Los Angeles County, Southern California*. [online] Available at: https://commons.wikimedia.org/wiki/File:Harbor_Freeway.svg [Accessed 11 January 2021]. Source adapted.]

Data mining techniques can be applied to the spatial data on the map to provide additional data analysis.

- (c) Explain how **two** data mining techniques could be applied to provide a different view of the data displayed on this map.

[6]

Markscheme

Award [6 max]

Apply a heat map filter (red for hot and yellow for cooler);

This shows the density of emergency calls;

The method used (e.g. Nature breaks, Quantile) can provide different maps;

Depending upon the colour key the maps can appear very different;

Filter by variable (e.g. type of emergency);

This would only show the dots for that emergency type;

This may show if there an outbreak in one area;

Emergency vehicles could be located near these areas;

There may be confounding data like population density;

Filter by time would show emergencies over a time period;

May be useful if an annual event (e.g. protest, concert) is due;

The data from the previous year can be analysed;

To assist with the deployment of emergency vehicles/increase police presence in those

areas to reduce emergencies;

Mark as [3] and [3].

Examiners report

The majority of students did not address the demands of the question. Most of them mentioned a data mining technique with some attempt at explaining in general terms how it could be applied to the data, but not really to the scenario; there was very little evidence to describe how the new data generated could change the view of the map.

Visual data analysis tools can be used to make the data on maps appear differently. For example, two different political parties may want the map data to show their different perspectives.

(d) Discuss the ethics of using visual data analysis tools.

[5]

Markscheme

Award [5 max]

Negative

Many people look at a map and believe that it is an objective truth;
They aren't aware that the data can be filtered to change the appearance;
Thus, map data should always include a key and other objective data;

Positive

If the agenda has a positive impact, then it is fine to use these tools;
e.g. show all crime (even minor crime) to boost the funding for emergency services;
May make it easier for decision makers to understand the data;
Which in turn, may save lives;

Conclusion

It is ethical to use all tools at your disposal. The opposition may also use these tools;
You run the risk that a counter map will be produced showing a very different perspective to yours;

Examiners report

Many students dwelt too much on the preamble of the question; the fact that visual data analysis tools can sometimes present a misleading view of data. They explained the ethics of this misrepresentation rather than proffering mitigation actions and outlining situations where the use of

these tools could be beneficial. No student got all the marks available for this question.

- (e) Describe how deviation detection can be used to predict potential incidents, such as fires.

[4]

Markscheme

Award [4 max]

Detection deviation is any usual pattern in the data/look for cases that are outside of the norm;

These deviations are compared against historical data over time;

e.g. Fires are clustered in one region;

e.g. Fires occur at certain times of the year;

The model should try to ignore random patterns (noise);

By setting a confidence level/e.g. 95%.

Examiners report

The majority of students showed understanding of deviation detection, but they could not demonstrate how it could be applied in the situation described. Most students focused on using observed deviations to predict fires rather than using them to improve the accuracy of prediction.